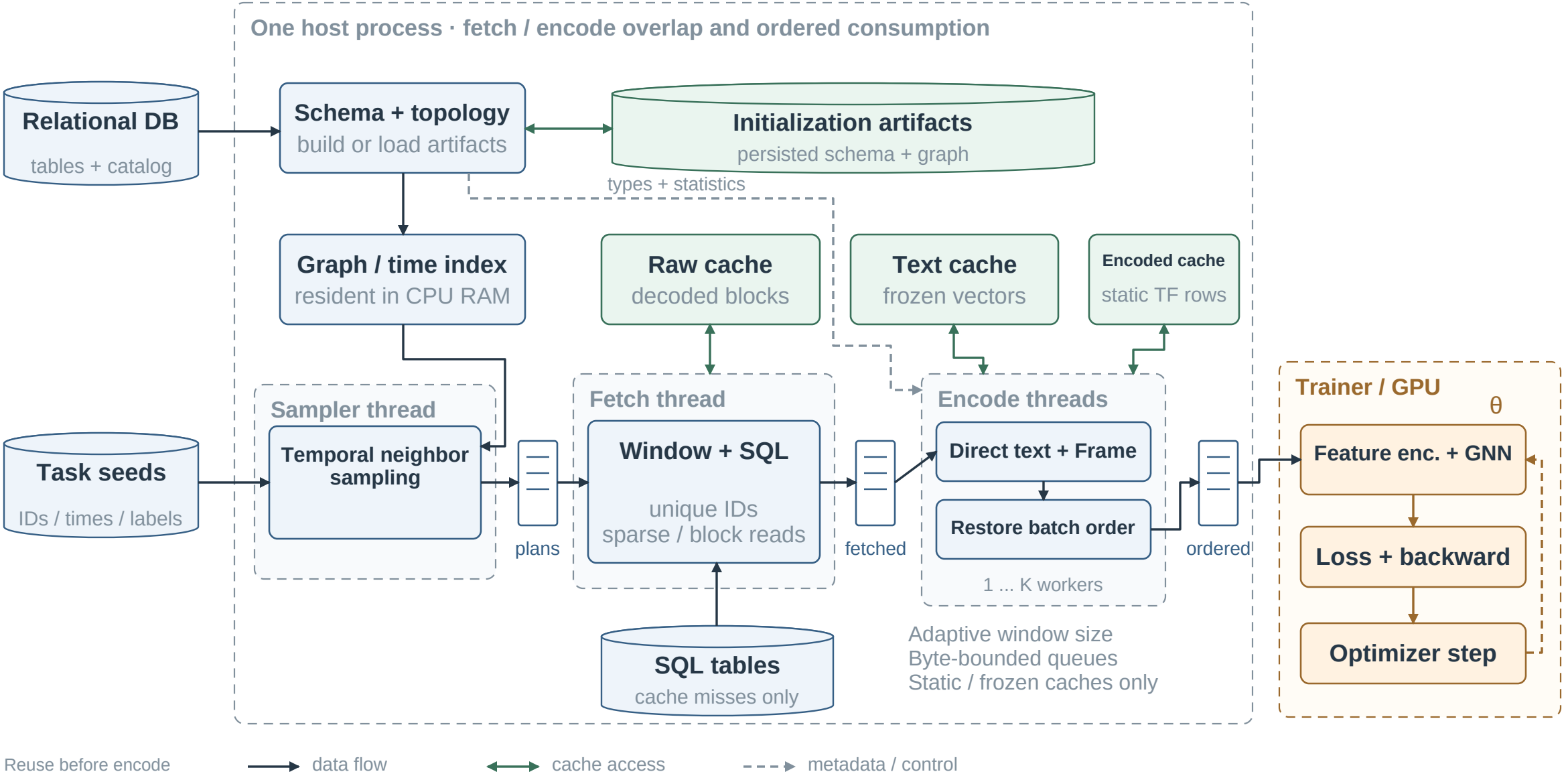
One host process · fetch / encode overlap and ordered consumption
Relational DB
tables + catalog
Schema + topology
build or load artifacts
Initialization artifacts
persisted schema + graph
types + statistics
Graph / time index
resident in CPU RAM
Raw cache
decoded blocks
Text cache
frozen vectors
Encoded cache
static TF rows
Task seeds
IDs / times / labels
Sampler thread
Temporal neighbor sampling
plans
Fetch thread
Window + SQL
unique IDs
sparse / block reads
fetched
Encode threads
Direct text + Frame
Restore batch order
1 ... K workers
ordered
SQL tables
cache misses only
Adaptive window size
Byte-bounded queues
Static / frozen caches only
Trainer / GPU
θ
Feature enc. + GNN
Loss + backward
Optimizer step
Reuse before encode
data flow
cache access
metadata / control